



DIGITAL FORENSICS IN NETWORKS-3332-SEC-3

Chapter 2: Fundamentals of Digital Evidence

Course Instructor

Dr. SUHAIL QADIR

Assistant Professor

Department: Information Systems

College: College of Computer Science

Email: sqadir@kku.edu.sa

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Fundamentals of Digital Evidence

Definition, characteristics, types, legal considerations, and collection principles

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Sources of Network-Based Evidence

The twelve places forensic investigators look across the network

03

Challenges of Network Evidence

Why network-based evidence is uniquely difficult to acquire and admit

Introduction

- Digital Evidence refers to any information or data of probative value that is stored, transmitted, or received in digital form.
- Found in devices such as computers, smartphones, IoT devices, servers, cloud systems, and across networks.
- It is intangible (not physical) but can be critical in legal proceedings.



Characteristics of Digital Evidence

Volatile

Easily altered, destroyed, or lost

e.g. RAM contents, network traffic

Intangible

Exists in binary form (0s and 1s).

Replicable

Can be copied without degradation

Time-sensitive

Certain evidence may only be available for a short duration.

Integrity Dependent

Any modification can challenge admissibility in court.

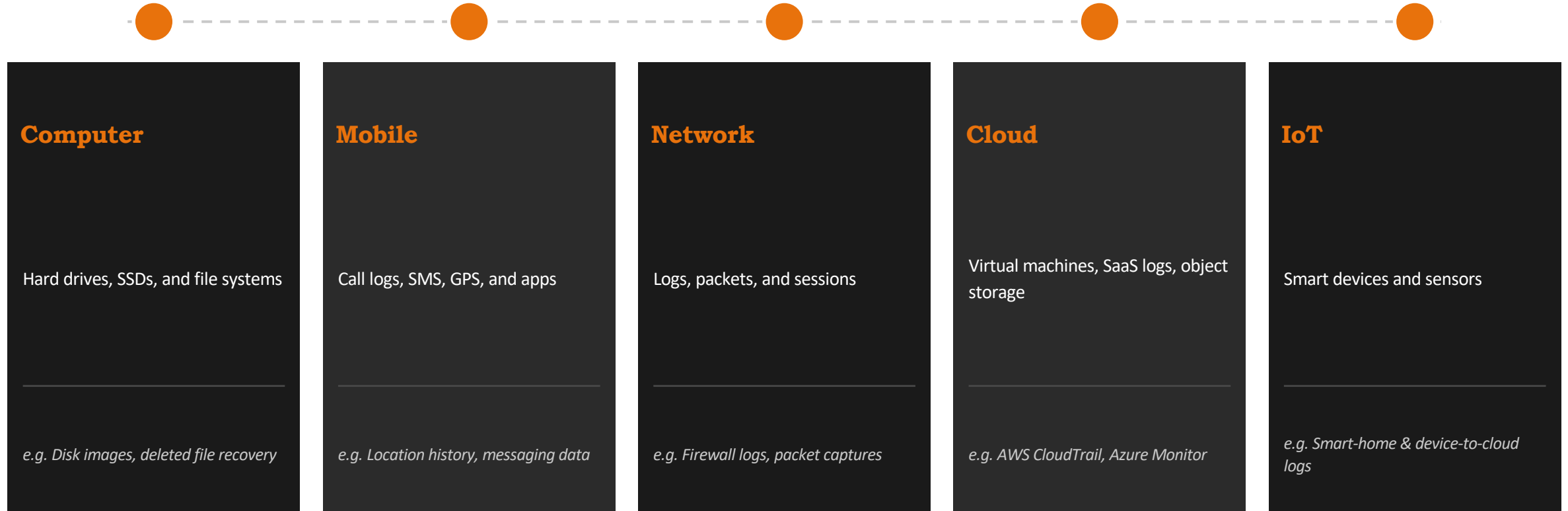
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Jurisdictional Complexity

Digital evidence often crosses borders (cloud, remote servers)

Types of Digital Evidence

Five categories of digital evidence, spanning devices, networks, and the cloud



Investigators must know where each category lives before it can be collected

Legal Considerations

Five requirements digital evidence must satisfy before a court will consider it



Admissibility

Must follow legal standards such as the Daubert standard and Federal Rules of Evidence



Authenticity

Must prove the evidence is genuinely what it claims to be



Chain of Custody

Documenting every step of collection, transfer, storage, and analysis



Integrity

Verified using hashing algorithms such as MD5, SHA-1, and SHA-256



Privacy & Jurisdiction

May involve international frameworks such as GDPR and HIPAA

Chain of Custody & Integrity



Chain of Custody

The documented trail proving who collected, handled, transferred, and stored a piece of evidence at every stage of an investigation.

Any gap in the record — an unexplained hand-off or missing timestamp — can be used to challenge the evidence's authenticity in court.



Integrity via Hashing

A cryptographic hash acts as a digital fingerprint. Recomputing it after collection proves the copy is bit-for-bit identical to the original.

- MD5 — fast, legacy checksum
- SHA-1 — improved, now deprecated
- SHA-256 — today's standard

Collection & Preservation Principles

Forensic soundness starts the moment an investigator touches a device

01

Follow Forensic Soundness

No alteration during collection — the original evidence must remain unchanged

02

Use Write-Blockers

Hardware or software write-blockers prevent any change while imaging drives

03

Document Every Step

Record metadata, timestamps, and investigator details throughout the process

04

Use Hash Values

MD5, SHA-1, or SHA-256 hashes prove that data integrity was preserved

05

Prefer Bit-Stream Imaging

A full bit-stream image is preferred over a logical copy of the data

Evidence Collection Methods

Four ways to duplicate digital media without altering the original

1

Image Copy

A straightforward file-level copy of the data

2

Bit-Stream Image

A complete sector-by-sector image of the entire drive

3

Bit-Stream Copy

A duplicate produced directly from the bit-stream image

4

Mirror Copy

A live, exact duplicate drive kept in sync with the original

An abstract network diagram consisting of several small grey circular nodes connected by thin, light grey lines. The lines form a complex web of connections across the dark grey background, with some lines intersecting and others forming closed loops.

PART 2

Sources of Network-Based Evidence

Twelve places where an investigator can find evidence — from the physical cable to the central log server

Where Evidence Lives

Every environment is unique — a large bank and a small clinic run very different equipment. Even so, most organizations share common categories of network infrastructure, and each category offers its own kind of evidence.

Overlapping

Sometimes multiple sources capture the same event — valuable for cross-correlation

Complementary

Other times each source holds a unique fragment of evidence found nowhere else

Uneven value

Usefulness depends on investigation type, device configuration, and network topology

The Twelve Sources

On the Wire	In the Air	Switches	Routers	DHCP Servers	Name Servers
Authentication Servers	NIDS / NIPS	Firewalls	Web Proxies	Application Servers	Central Log Servers

The Physical Layer: On the Wire & In the Air

1

On the Wire

Copper (twisted pair / coaxial) or fiber-optic cabling connects stations to switches and switches to routers, carrying signals as voltage changes or pulses of light.

FORENSIC VALUE

Investigators can tap physical cabling to copy and preserve traffic — from crude “vampire” taps to purpose-built infrastructure taps that replicate signals without degrading the original.

2

In the Air

Wireless networking (RF, occasionally IR) lets stations communicate without line-of-sight, since radio waves pass through air, wood, and brick with some signal loss.

FORENSIC VALUE

Investigators can capture wireless frames to identify legitimate vs. suspicious stations by MAC address, run volume-based traffic analysis, and inspect full packets when traffic is unencrypted.

Switches & Routers

3 Switches

Multiport bridges that physically connect stations or segments to form a LAN — “core” switches aggregate traffic, “edge” switches connect individual stations.

Forensic value: a switch's CAM table maps MAC addresses to physical ports, and a “mirror” port lets investigators capture live traffic with a sniffer.

4 Routers

Connect different subnets or networks and route packets between segments, even across different addressing schemes — the foundation of the global Internet.

Switch — CAM Table

Port	MAC Address
Gi0/1	3C:5A:B4:11:9F:02
Gi0/2	A0:CE:C8:77:2D:6B
Gi0/3	F4:5C:89:00:1A:3E

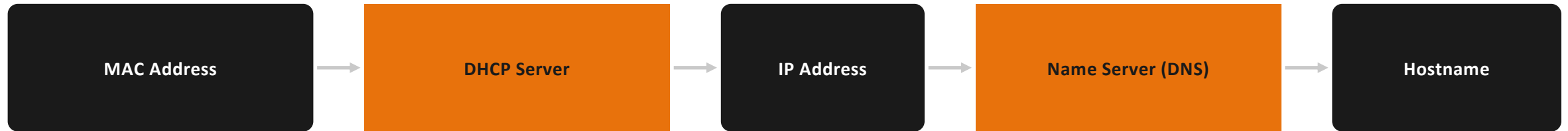
Router — Routing Table

Destination	Next Hop
10.30.0.0/16	192.168.1.1
172.16.0.0/12	192.168.1.254
0.0.0.0/0	203.0.113.1

Routers may also log denied traffic and export flow records to a central log server for correlation.

DHCP Servers & Name Servers

Two mapping services that translate between machine and human-readable identities



DHCP Servers

Dynamically assign IP addresses to LAN stations rather than requiring administrators to configure static addresses manually. Often built into routers or wireless access points.

Forensic value: DHCP lease logs record the assigned IP, the requesting MAC address, and the lease time — letting an investigator pinpoint exactly which device held an IP address at a given moment.

Name Servers (DNS)

Just as DHCP maps MAC to IP, the Domain Name System maps IP addresses to the human-readable names we assign to systems and networks, via queries against central DNS servers.

Forensic value: DNS query logs can reveal which hostnames a device resolved — often surfacing command-and-control domains or exfiltration destinations.

Authentication Servers & NIDS / NIPS

5

Authentication Servers

Provide centralized login services so user accounts are managed in one place instead of on thousands of individual machines, streamlining provisioning and auditing.

FORENSIC VALUE

Logs of successful and failed logins reveal brute-force attacks, logins at suspicious hours or locations, and unexpected privileged access — across the entire authentication domain.

6

NIDS / NIPS

Network intrusion detection (and prevention) systems monitor traffic in real time for signs of adverse events, using signature, protocol-awareness, or behavioral analysis.

FORENSIC VALUE

Purpose-built to alert on security events: attacks in progress, command-and-control traffic, or misconfigurations — value depends heavily on sensor placement and coverage.

Firewalls & Web Proxies

7

Firewalls

Filter traffic based on IP addresses, ports, payloads, and encapsulated protocols. Deployed at network perimeters and increasingly within internal networks to isolate segments.

FORENSIC VALUE

Modern firewalls have granular logging: allowed/denied traffic, configuration changes, and errors — functioning as both protection devices and a useful pseudo-IDS.

8

Web Proxies

Sit between clients and the web to improve performance through caching, and to log, inspect, and filter web traffic — often against blacklists or content keywords.

FORENSIC VALUE

Logs can tie web activity to a specific user (via directory correlation), and cached pages can sometimes be retrieved or carved for evidence of malware.

Application Servers & Central Log Servers

9

Application Servers

Every organization runs a mix of application servers, varying by industry, size, and mission — each a potential evidence source in its own right:

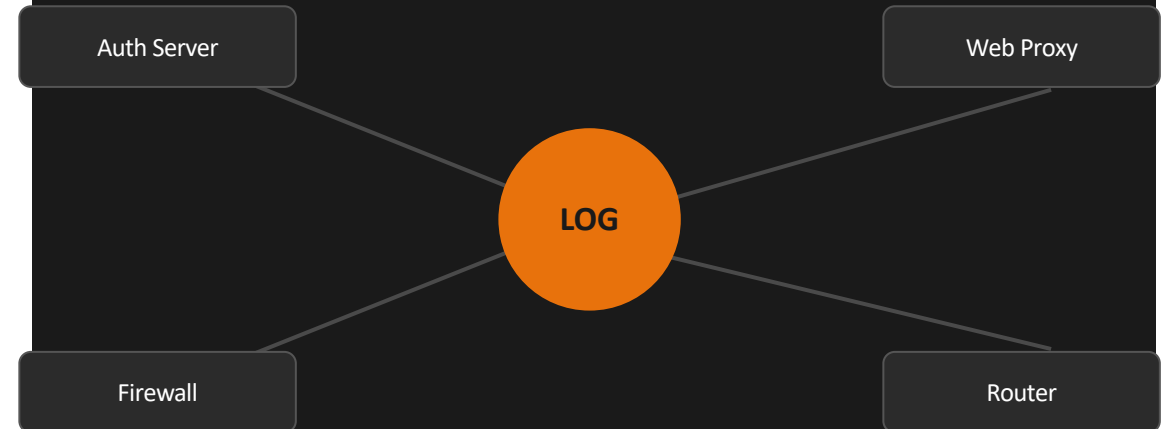
- Database servers
- Web servers
- Email servers
- Chat servers
- VoIP / voicemail servers

Review network diagrams and application documentation to identify which servers matter most for your investigation.

10

Central Log Servers

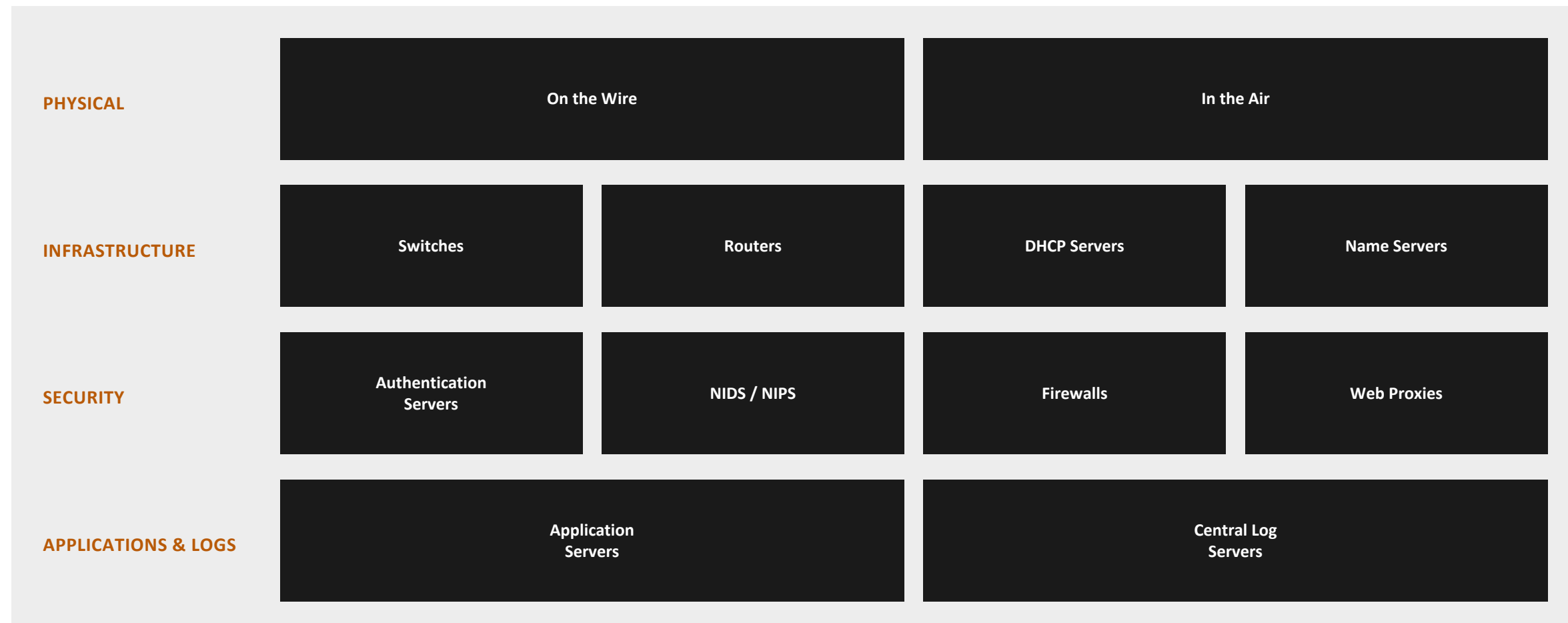
Aggregate event logs from many sources — timestamped, correlated, and analyzed far more easily than logs scattered across disparate systems.



Even a compromised or rebooted device may leave its logs intact on the central server, retained far longer than the original allows.

The Network Evidence Map

All twelve sources, organized by where they sit in a typical network environment



Challenges of Network-Based Evidence

Network evidence poses special difficulties across five dimensions of an investigation



Encryption

TLS/SSL, VPNs, and end-to-end encryption make traffic unreadable without the keys



High Volume

Terabytes of traffic demand filtering and automation to make review feasible



Volatility

Network traffic disappears quickly if it isn't captured live



Attribution

Linking an IP address to an individual is hard with NAT and shared devices



Jurisdiction

Cross-border investigations complicate evidence gathering and legal authority

Thank You

Questions & discussion welcome.